

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

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Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

1. Aim and Objective

- To implement a **Decision Tree classifier**.
- To extend the Decision Tree into a **Random Forest ensemble model**.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2. Dataset Description

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: <https://archive.ics.uci.edu>

3. Preprocessing Steps

Before training the regression models, the dataset was preprocessed to improve data quality and ensure compatibility with the algorithms.

Loading the Dataset

The Breast Cancer Wisconsin (Diagnostic) dataset was loaded using the `ucimlrepo` library. This dataset is widely used for binary classification in medical diagnosis.

- Number of instances: 569
- Number of input features: 30 numerical features
- Target variable: Diagnosis (M = Malignant, B = Benign)

The features describe characteristics of cell nuclei extracted from breast mass images, such as radius, texture, area, smoothness, and concavity.

The objective is to build a model that classifies tumors as malignant or benign based on these features.

Encoding of Class Labels

The target variable contains categorical labels: **M** (Malignant) and **B** (Benign). Since machine learning algorithms require numerical input, the class labels were encoded using `LabelEncoder`.

- Benign (B) $\rightarrow$ 0
- Malignant (M) $\rightarrow$ 1

This transformation converts categorical values into binary numerical form suitable for classification models.

Class Label Visualization

To understand the distribution of classes, a count plot was generated. The visualization shows the number of benign and malignant cases in the dataset.

This helps identify whether the dataset is balanced or imbalanced, which is important for model evaluation.

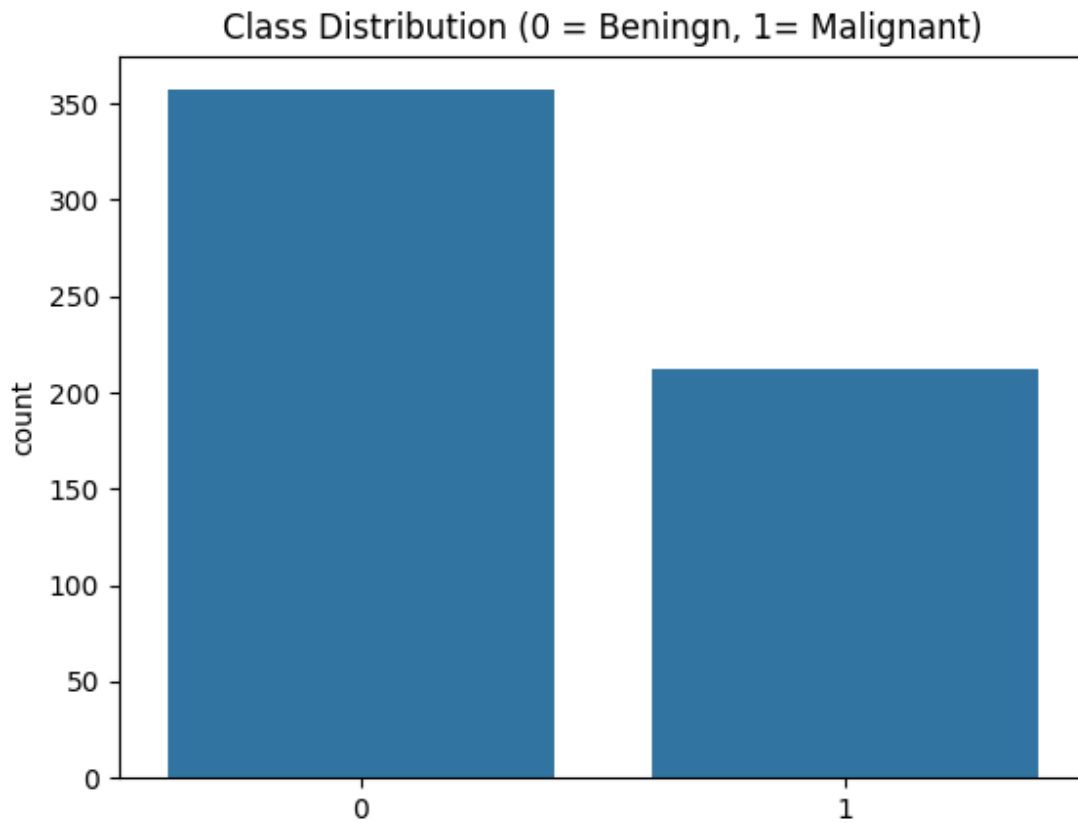


Figure 1: Distribution of Loan Sanction Amount

Feature Correlation Analysis

A correlation heatmap was generated to examine relationships between numerical features.

The heatmap displays the Pearson correlation coefficients between feature pairs, helping to identify:

- Highly correlated features
- Redundant variables
- Strong relationships that may influence model performance

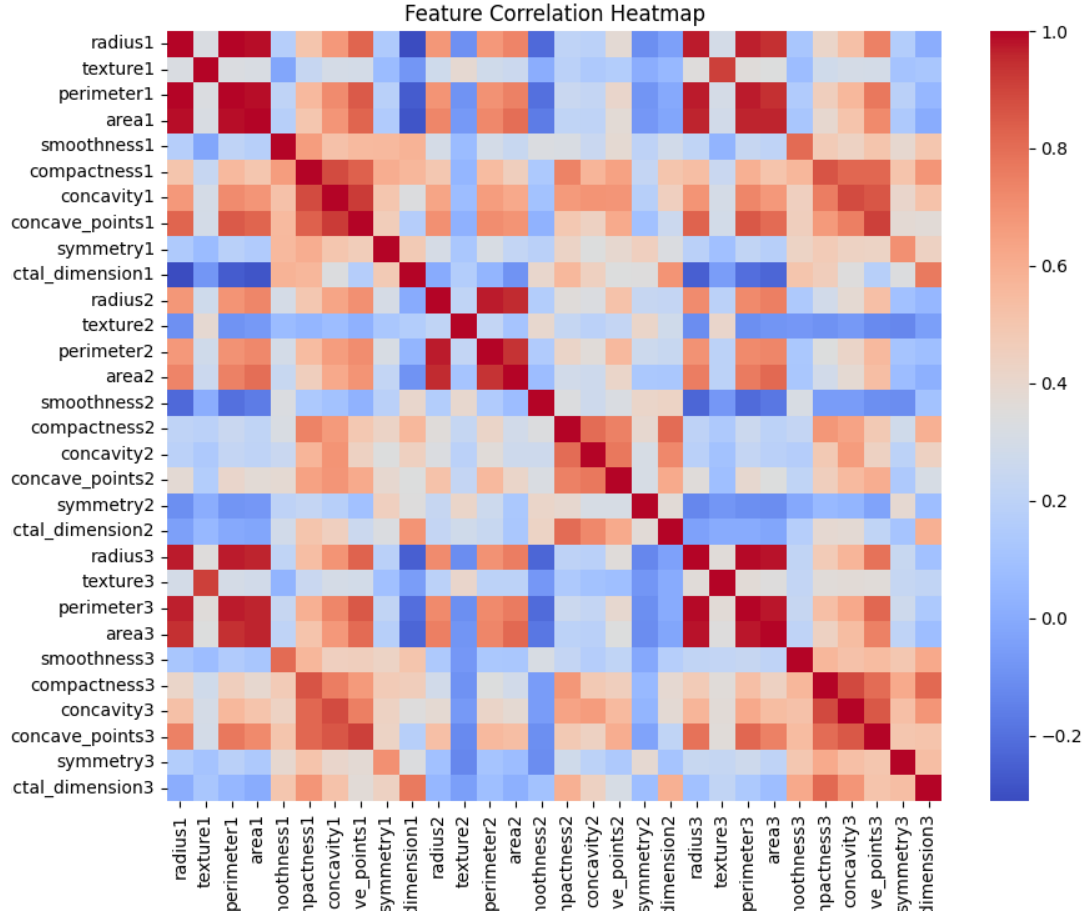


Figure 2: Correlation Heatmap of Breast Cancer Dataset Features

Understanding feature correlation assists in feature selection and improving model efficiency.

Train-Test Split

The dataset was split into training and testing sets to evaluate model performance.

- Training set: 80% of the data
- Testing set: 20% of the data

This split ensures that model evaluation is performed on unseen data.

Result of Preprocessing

After completing the preprocessing steps, the dataset was clean, numerically encoded, and properly scaled, making it suitable for training classification models.

4. Implementation Details

Decision Tree Implementation

A Decision Tree classifier was implemented using `DecisionTreeClassifier`. The model was trained on the training dataset and tested on unseen test data.

Performance metrics used:

- Accuracy
- Precision
- Recall
- F1-score

The initial Decision Tree achieved approximately 93% accuracy on the test set.

Hyperparameter Tuning

To improve model performance, hyperparameter tuning was performed using `GridSearchCV` with 5-fold cross-validation.

The following parameters were optimized:

- `max_depth`
- `min_samples_split`
- `min_samples_leaf`
- `criterion` (gini, entropy)
- `max_features`

The best model was selected based on cross-validation accuracy and F1-score. After tuning, the Decision Tree accuracy improved to approximately 95.6%.

Random Forest Implementation

A Random Forest classifier was also implemented using `RandomForestClassifier`. Hyperparameter tuning was performed using `GridSearchCV` to optimize:

- `n_estimators`
- `max_depth`
- `min_samples_split`
- `min_samples_leaf`
- `max_features`

The tuned Random Forest achieved approximately 97.3% test accuracy, outperforming the Decision Tree model.

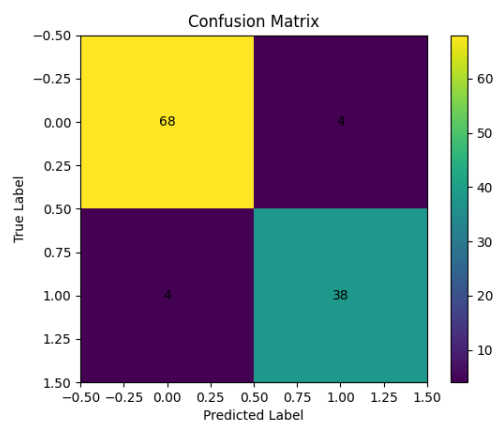
Model Evaluation and Comparison

Both models were evaluated using:

- Confusion Matrix
- 5-Fold Cross-Validation Accuracy
- ROC Curve
- Area Under Curve (AUC)

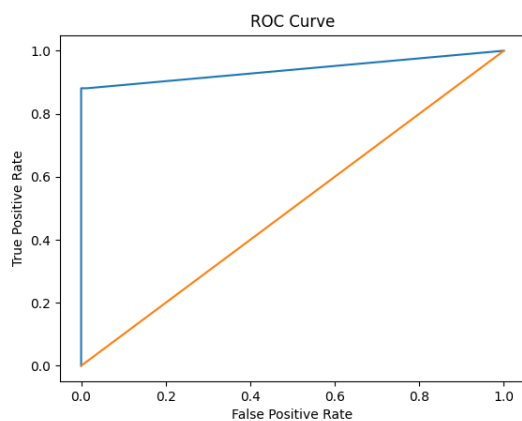
The Random Forest model showed superior performance in terms of accuracy, F1-score, and AUC score, making it the better-performing model for this classification task.

The following visualizations were generated to analyze data distribution and model behavior:

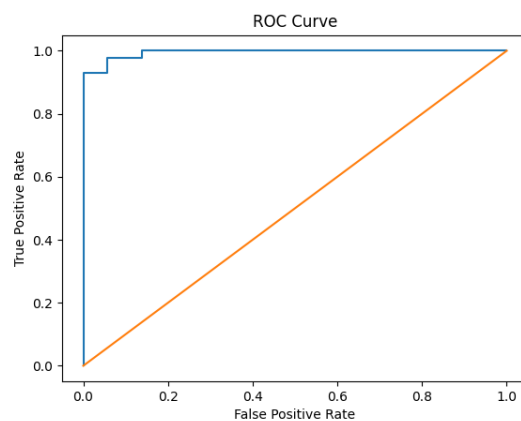


(a) Distribution of Loan Sanction Amount

Figure 3: Exploratory Data Analysis Visualizations



(a) Relationship Between Applicant Income and Loan Amount



(b) Relationship Between Applicant Income and Loan Amount

Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
entropy	None	94.73	0.9297
entropy	20	94.73	0.9297
entropy	10	94.73	0.9297

Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	None	sqrt	96.70	0.9561
50	10	sqrt	96.70	0.9561
50	20	sqrt	96.70	0.9561

5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	94.7	93.9	95.6	93.0	96.5	94.73
Random Forest	96.5	95.6	98.2	94.7	98.2	96.70

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve and AUC

Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Hyperparameters were selected based on average cross-validation performance, ensuring

robust generalization. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree.

References

- Scikit-learn: Decision Trees
- Scikit-learn: Random Forest
- UCI Dataset